

$$\textcircled{1} \quad A = \begin{pmatrix} 4 & -1 & -1 \\ -1 & 4 & -1 \\ -1 & 0 & 2 \end{pmatrix} = D - E - F$$

Og 35

a) Jacobi:  $B_J, \rho(B_J)$

b)  $\omega$  (relax, Jacobi),  $\omega_{opt}, \rho(B_{J\omega_{opt}})$

a)  $B_J = D^{-1}(E + F)$

$$= \begin{pmatrix} 4 & & \\ & 4 & \\ & & 2 \end{pmatrix}^{-1} \underbrace{\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}}_{E+F}$$

$$= \begin{pmatrix} \frac{1}{4} & & \\ & \frac{1}{4} & \\ & & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{2} & 0 & 0 \end{pmatrix}$$

$$B_J = \begin{pmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{2} \end{pmatrix} \underbrace{\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}}_C$$

$$0 = \det(B_J - \lambda I)$$

$$0 = \det(C - \mu I)$$

$$= \det \begin{pmatrix} -\mu & 1 & 1 \\ 1 & -\mu & 1 \\ 2 & 0 & -\mu \end{pmatrix}$$

$$= 2 \det \begin{pmatrix} 1 & 1 \\ -\mu & 1 \end{pmatrix} - \mu \det \begin{pmatrix} -\mu & 1 \\ 1 & -\mu \end{pmatrix}$$

$$= 2(1 + \mu) - \mu(\mu^2 - 1)$$

$$= 2(1 + \mu) - \mu(\mu - 1)(\mu + 1)$$

$$= \underbrace{(\mu + 1)} \underbrace{(2 - \mu(\mu - 1))}$$

$$\underline{\mu_1 = -1}$$

$$2 - \mu^2 + \mu = 0$$

$$\mu^2 - \mu - 2 = 0$$

$$(\mu - 2)(\mu + 1) = 0$$

$$\mu_2 = \underline{-1}$$

$$\mu_3 = \underline{2}$$

$\mu_i$  EW van  $C$

$$\text{EW } B_2: \underline{\lambda_{1,2} = -\frac{1}{4}}, \underline{\lambda_3 = \frac{1}{2}}$$

$$\Rightarrow \rho(B_2) = \frac{1}{2}$$

$$b) \quad \mathcal{I} \omega : \quad \mathcal{B}_2 : \text{di null}$$

$$-1 < \lambda_1 \leq \dots \leq \lambda_n < 1$$

$$\omega_{\text{opt}} = \frac{2}{2 - \lambda_1 - \lambda_n}$$

$$= \frac{2}{2 - (-\frac{1}{4}) - \frac{1}{2}} = \frac{8}{7}$$

$$\lambda_{i, \omega_{\text{opt}}} = 1 - \omega_{\text{opt}} + \omega_{\text{opt}} \lambda_i$$

$$\uparrow$$
  

$$\mathcal{B}_2$$

$$\lambda_{1,2, \omega_{\text{opt}}} = -\frac{3}{7} \quad \Bigg|$$

$$\lambda_{3, \omega_{\text{opt}}} = \frac{3}{7}$$

$$\rho(\mathcal{B}_{\mathcal{I} \omega_{\text{opt}}}) = \frac{3}{7}$$

$$\rho(\mathcal{B}_2) = \frac{1}{2}$$